



SCHOOL OF BUSINESS

MASTER'S DEGREE IN BUSINESS INTELLIGENCE AND DATA SCIENCE

Network Analytics Model for Identifying Concentration Patterns and Recurrent Relationships in Ecuadorian Public Procurement

Analysis of Electronic Reverse Auction Procedures During 2025

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Executive Summary

The aim of the study was to analyse the relations between contracting entities and suppliers in the procedures of Electronic Reverse Auction initiated in Ecuador during 2025, in order to identify patterns of concentration, recurrence and structure useful for prioritizing reviews. Open SERCOP data were used, which were cleaned and integrated through OCID. The analysis combined descriptive statistics, Spearman correlation and concentration indicators - standardized HHI, CR4 and entropy - with a bipartite Buyer-Supplier network model built in Neo4j. Connectivity, centrality and communities were examined through WCC, degree, betweenness centrality, Louvain and Leiden; in addition, recurrences were defined as more than three procedures for the same combination entity and the provider-CPC-5 within 90 days. The results showed that 52.8% of the procedures had a maximum of three bidders and that the network was composed of 13,493 relationships. Louvain and Leiden revealed a consistent community structure, with modularity close to 0.69, and made it possible to visualize sectoral communities, including one consisting mainly of hospitals linked by shared suppliers. The combination of signals of amount, recurrence, single-bidder participation, concentration and betweenness centrality reduced the universe analyzed to 71 relationships with three or more signals and 3 with four. It is concluded that network analysis complements tabular analysis by revealing unobvious connections and clusters, and provides SERCOP with a reproducible mechanism for prioritizing. However, the results are descriptive: they guide the technical and documentary review, but they do not constitute evidence of irregularity.

Abstract

The objective of this study was to analyze the relationships between contracting entities and suppliers in Electronic Reverse Auction procedures initiated in Ecuador during 2025, in order to identify patterns of concentration, recurrence, and network structure that could support the prioritization of reviews. Open data from SERCOP were cleaned and integrated using the OCID. The analysis combined descriptive statistics, Spearman's rank correlation, and concentration measures—normalized HHI, CR4, and entropy—with a Buyer-Supplier bipartite network built in Neo4j. Network connectivity, centrality, and communities were examined using WCC, degree, betweenness centrality, Louvain, and Leiden. Recurrence was defined as more than three procedures involving the same contracting entity-supplier-CPC-5 combination within a 90-day period. The results showed that 52.8% of the procedures had no more than three bidders and that the network comprised 13,493 relationships. Louvain and Leiden revealed a consistent community structure, with modularity close to 0.69, and made it possible to identify sector-specific communities, including one composed mainly of hospitals connected through shared suppliers. Combining signals related to awarded amount, recurrence, single-bidder participation, concentration, and intermediary position reduced the analyzed universe to 71 relationships with three or more signals, of which 3 presented four signals. The study concludes that network analytics complements traditional tabular analysis by revealing connections and clusters that would otherwise remain difficult to detect, providing SERCOP with a reproducible mechanism for prioritizing reviews. However, the findings are descriptive: they guide technical and documentary review but do not constitute evidence of irregularities.

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1. Introduction

The Ecuadorian State is conducting numerous procurement procedures to purchase the goods and services needed by its institutions. As a result, records are generated on public buyers, bidders, awarded suppliers, economic values and dates. The National Public Procurement Service (SERCOP, 2026) facilitates access to this information through its open contracting platform. However, having the files does not mean that the relationships present in them can be identified immediately.

When each procedure is reviewed individually, it is difficult to recognize which entities contract with the same suppliers, which links are repeated over a given period or which actors occupy relevant positions within the market. It is also difficult to establish whether contracts are awarded among several suppliers or whether a significant part of their economic value is concentrated in few. For this reason, the research proposes to organize the data as a network for joint observation of the connections.

Newman (2003) defines networks as structures formed by actors and relationships. Sayama (2015) complements this approach by explaining that the functioning of a system depends on how its members interact. On the basis of these grounds, the contracting entities and the suppliers awarded were represented as nodes, while the awards were used to establish the connections between them. This structure is called the Buyer-Supplier bipartite network because it relates two different types of actors: buyers and suppliers.

The investigation includes the procedures of Electronic Reverse Auction that began in Ecuador between January and December 2025. Twelve monthly files published by SERCOP were collected and three databases were organized. The first contains 18,326 procedures; the second, 79,425 records of participation of bidders; and the third, 15,718 awards.

After linking the buyers to the awarded suppliers, the final network was made up of 7,664 actors and 13,493 relationships.

The study uses a descriptive and relational approach, because its purpose is to understand the organization of the network and not to predict whether a procedure is regular or irregular. Python was used to integrate the files, review the quality of the records, treat the missing data and build the necessary variables. Neo4j Graph was later used to represent the network and calculate its structural characteristics.

The analysis included the number of connections of each actor, the frequency and amount of relationships, the centrality of betweenness centrality and the formation of communities through the Louvain and Leiden algorithms. The concentration of awards, the repetition of relations within 90 days and the proportion of single-bidder procedures were also studied. Burgos Carlosama (2025) warns that the amount of awards and their economic value should be examined separately, because a frequent relationship does not necessarily concentrate the higher amounts.

The contribution of the project is not limited to the presentation of rankings of entities or suppliers, but it allows to observe how public procurement is organized through the full

network, selected subnetworks, communities and patterns of shared suppliers. Among the main results, a central component was identified that brings together 92.6% of the actors, as well as groups of buyers and suppliers with similar internal connections. Important differences were also found between the concentration by amount and frequency, in addition to recurrent relationships within 90 days. Finally, the combination of five signals reduced the 13,493 network relationships to 71 with three or more signals and to 3 with four signals. These results provide a clearer basis for directing the review to a specific group of relationships. However, they should be interpreted as priority criteria and not as direct evidence of fraud, collusion or irregularity.

2. Revision Of Literature

To analyse public procurement it is not enough to review separately the number of procedures, the suppliers awarded or the awarded amounts. It is also necessary to note how the participants relate, how often these links are repeated and what position each actor occupies within the system.

Since network science, Newman (2003) proposes to study systems from their nodes and the connections between them. Sayama (2015) expands this idea by pointing out that the behavior of a system arises not only from the characteristics of its components, but also from the way they interact. Applicable to this investigation, contracting entities and suppliers can be represented as nodes, while the awards make up the links of the network. This structure will allow to examine the recurrence of relationships, the associated amounts and the presence of actors with a prominent participation.

Studies in Ecuador show that network analysis allows information that does not always appear in traditional descriptive methods. Burgos Carlosama (2025) found that more relationships do not necessarily involve greater economic concentration. Therefore, the frequency of the awards and the awarded amounts must be analysed jointly, but differentiated.

Redroban Ortiz et al. (2024) showed that network metrics, such as centrality, help to identify entities and suppliers with high participation or recurrent connections. However, these results should be interpreted as signs for further analysis and not as direct evidence of irregularity.

International evidence also shows that the concentration varies from market to market. Pliatsidis (2024) identified sectors in Greece dominated by few suppliers and others with a more balanced distribution. This indicates that the

concentration should be assessed by considering the type of good or service, the number of competitors and the conditions specific to each sector.

Competition also has an impact on the structure of the procurement network. Jamett (2022) found that 20.21% of the tenders awarded in Chile between 2019 and 2021 had one bidder. The author related this low participation to the complexity of the bases, documentary

requirements and other entry barriers. This shows that an electronic platform alone does not guarantee wide competition.

Lyra et al. (2021) analyzed the joint participation of suppliers in municipal tenders in Ceará, Brazil. Through a network of companies, they identified 22 communities with similar patterns of specialization, activity and territorial scope. The study shows that recurrent relationships can reveal structures that are not observed when reviewing the records in isolation.

Wachs et al. (2021) point out that the percentage of contracts awarded is not sufficient to understand the structure of the market. It is also necessary to analyse the position of each actor, its connections and its membership in communities. These patterns should be understood as signs for further analysis and not as direct evidence of irregularity.

Data quality is essential for this type of study. López Molina (2022) recognized the progress of SERCOP in the publication of open data, but also identified problems of standardization, organization and understanding. These limitations can generate duplication and affect the correct construction of the network.

Guerrero Cisneros (2025) showed that data from Electronic Reverse Auction can be used in advanced computational models. This antecedent confirms its analytical value, but also highlights the need to purify, normalize and validate the information before building the network.

In conclusion, the revised literature makes it possible to understand public procurement as a network of relationships between entities and suppliers, and not just as a set of isolated processes. Network analysis facilitates the identification of relevant actors, recurrent links, communities and concentration levels, although these results should be interpreted according to market characteristics, competition and data quality.

Despite the progress made, no study was found to analyse in a comprehensive way the Electronic Reverse Auction in Ecuador during 2025, relating the recurrence of the links, the awards and the awarded amounts. This research therefore proposes to develop in Neo4j a model of graphs that will enable you to visualize the connections between entities and suppliers, identify concentration patterns and generate information useful for public monitoring and decision-making.

Table 1A Similar studies - research design

Author and year	Study	Data	Methodology
Burgos Carlosama (2025)	Network analysis in public procurement: an approach to actor interactions in the acquisition of medical supplies and school food in Ecuador (2020-2021)	Official SERCOP open data on buyers, suppliers, awards, contracts, tenders, OCIDs, and awarded amounts.	Directed-network analysis. Buyers and suppliers as nodes; awards as relationships; awarded amount as weight. R, RStudio, and Gephi.
Redroban Ortiz et al. (2024)	Complex networks as a methodology for government control of public procurement processes in Ecuador	19,797 SERCOP procurement procedures from 2023, including suppliers, contracting entities, procurement method, tax ID, and procedure code.	Complex-network analysis in Cytoscape 3.10.1 using degree, betweenness, closeness, density, and centralization.
López Molina (2022)	Implementation of open data	Open data from the Economic	Qualitative and inductive

Author and year	Study	Data	Methodology
	and its contribution to transparency and access to public information in Ecuador through 2021	and Productive Sector Cabinet ministries and SERCOP, supplemented by interviews and observation of institutional websites.	approach using interviews, ethnographic observation, institutional review, purposive sampling, and action research.
Guerrero Cisneros (2025)	Application of machine-learning models to identify an early risk indicator in Ecuadorian Electronic Reverse Auction procedures	5,710 procedures from 2020, 2022, and 2023, including amounts, bidders, entities, process status, and contractual documents.	Machine learning and natural-language processing. Compared Random Forest, SVM, Gradient Boosting, and neural networks; combined structured variables with Doc2Vec representations.
Pliatsidis (2024)	Analyzing Concentration in the Greek Public Procurement Market: A Network Theory Approach	Greek public procurement open data from 2019-2021, organized by contracting entities, suppliers, awards, and CPV codes.	Directed networks and degree-distribution analysis to measure sector concentration.
Wachs, Fazekas, and Kertész (2021)	Corruption Risk in Contracting Markets: A Network Science Perspective	More than four million European public contracts from 2008-2016, including buyers, suppliers, sectors, and bidder counts.	Bipartite networks, core-periphery analysis, community detection, and single-bidder awards as a risk indicator.
Jamett (2022)	Competition and transparency in Chile's Public Market Information System	Open data from 285,252 awarded tenders in Chile during 2019-2021, supplemented by buyer, supplier, and specialist interviews.	Mixed approach combining descriptive statistics and semi-structured interviews.
Lyra et al. (2021)	Characterization of the Firm-Firm Public Procurement Co-Bidding Network from the State of Ceará Municipalities	196,608 participation records involving 45,502 firms in 84,835 municipal tenders in Ceará, Brazil, during 2015-2019.	Firm-to-firm network using Jaccard similarity, Louvain community analysis, and territorial and sectoral diversity metrics.

Table 1B Similar studies - findings and implications

Author and year	Main findings	Managerial implications
Burgos Carlosama (2025)	Found centralized school-food networks and broader medical-supply networks. Identified dominant actors by relationship count and awarded amount.	Supports detection of concentration, supplier dependence, and relevant relationships, and informs oversight, transparency, and public decisions.
Redroban Ortiz et al. (2024)	Electronic Reverse Auction showed the highest interaction. Suppliers and entities with high participation and recurrent relationships were also identified.	Can prioritize audits, detect concentration, and direct reviews toward highly recurrent actors. High connectivity is an alert signal, not proof of corruption.
López Molina (2022)	Ecuador advanced in open-data publication, with SERCOP among the most developed institutions; standardization, comprehension, and citizen-participation problems remained.	Formats should be standardized, user guidance improved, and institutional training and control strengthened so open data become useful management information.
Guerrero Cisneros (2025)	Binary classification better distinguished high- and low-risk procedures. Structured variables had greater predictive capacity than text used independently.	May support early risk-signal detection and prioritization for preventive review. Results are analytical alerts, not definitive evidence of corruption.
Pliatsidis (2024)	Identified sectors with high contract concentration among few suppliers and others with greater competitive balance.	Supports prioritizing sectors for oversight, strengthening competition, and informing control decisions.
Wachs, Fazekas, and Kertész (2021)	More centralized markets showed greater risk, although its distribution	Supports targeting controls and investigations according to the

Author and year	Main findings	Managerial implications
	varied between each country's core and periphery.	location and concentration of risk signals.
Jamett (2022)	20.21% of tenders had one bidder and 64.79% had between one and four bidders, demonstrating low competition.	Proposes simplifying and standardizing databases, reducing barriers for SMEs, and improving access to contractual information.
Lyra et al. (2021)	Identified 22 firm communities with similar participation, specialization, and geographic-reach patterns.	Supports prioritizing groups with atypical behavior for further oversight and analysis without treating the patterns as conclusive evidence of irregularity.

Note. *Own-production from the research mentioned in the table.*

2.1. Primary sources

The main information of this research will come from the official records published by the National Public Procurement Service (SERCOP) on the Equator-OCSS Open Procurement platform. The monthly files of the Electronic Reverse Auction procedures initiated in 2025 will be used.

Such records shall include data such as the contracting entity, the awarded supplier, the procedural code, the dates, the status of the process, the number of participants and the amount awarded. This information will make it possible to observe not only how many contracts were made, but also what actors were involved and how often their relationships were repeated.

Before analysis, the files will be integrated, cleaned and standardized to correct duplicates, incomplete data and differences in names or identifiers. The information will then be incorporated in Neo4j, where entities and suppliers will be represented as nodes, and awards as relationships. In this way, it will be possible to visualize the structure of the market and identify relevant actors, recurrent links and concentration patterns.

2.2. Secondary sources

The secondary sources will be composed of scientific articles, theses, academic books and technical reports related to public procurement, competition, the concentration of awards and the analysis of networks. The work of Newman (2003) and Sayama (2015) will serve as a theoretical basis for understanding the structure of the networks and the relationships between their actors.

The empirical background will be considered the investigations of Burgos Carlosama (2025), Redroban Ortiz et al. (2024), Pliatsidis (2024), Jamett (2022), Lyra et al. (2021), Wachs et al. (2021), López Molina (2022) and Guerrero Cisneros (2025). These studies will make it possible to compare the data used, the methodologies applied, the results obtained and their contributions to governance.

3. Identification Of The Study Object

The object of the study includes the relations between the contracting entities and the suppliers awarded in the procedures of the Electronic Reverse Auction initiated in Ecuador during 2025. The analysis was not limited to determining how many procedures were carried out or what their amounts were, but sought to understand how the actors are connected, how often their relationships are repeated and what participation each one reaches within the public procurement system.

The SERCOP publishes information on buyers, bidders, suppliers, awards and amounts. However, these data are distributed in monthly files and present each procedure separately. When the information is reviewed only by general tables or indicators, it is complex to recognize recurrent relationships, central actors, communities of buyers and suppliers or differences between monetary concentration and frequency concentration. The main organizational problem, therefore, is not the absence of data, but the difficulty in linking them and making them useful information for monitoring.

The literature shows that network analytics allow this limitation to be addressed. Burgos Carlosama (2025) points out that the number of awards and the amount contracted represent different dimensions and should therefore be considered separately. Lyra et al. (2021) shows that communities and repeated links allow for the discovery of structures that are not observed when analyzing each procedure individually. Similarly, Redroban Ortiz et al. (2024) explain that centrality metrics help to recognize actors with relevant positions within public procurement networks.

In order to respond to this need, a two-party buyer network was developed in Neo4j, in which the contracting entities and the suppliers were represented as nodes and the awards as relationships. On this structure, components, centrality, betweenness centrality, communities, concentration, recurrence and participation of

bidders. In this way, the project went beyond accounting for procedures and made it possible to study how the network is organized and what relationships have characteristics that justify a more detailed review.

The practical importance of the model lies in its ability to reduce a wide universe of information to an explicable group of priority relationships. Of the 13,493 Buyer-Supplier relationships built, 71 gathered three or more signs and 3 reached four. This focus can help to make more efficient use of monitoring resources, because it allows the review to be initiated with a reduced set of relationships and to know the reasons why they were selected.

The model does not replace the technical, documentary or legal review of the procedures. The concentration, recurrence, centrality and low participation of bidders are signs that guide the analysis, but do not alone demonstrate fraud, collusion or irregularity. In addition, the methodology can be updated with information from new periods and

adapted to other procurement modalities, provided that its variables, analysis units and thresholds are rerevised.

4. Problem Statement

The National Service for Public Procurement (SERCOP) makes available records of the procedures of Electronic Reverse Auction which contain information on contracting entities, suppliers, offers, awards and contracted amounts. Although these data allow each procedure to be consulted, they are usually distributed in extensive files and organized in tables that present the operations individually. As a result, its conventional review does not allow for easy observation of the complete structure of the relationships between entities and suppliers.

The organizational problem therefore lies not in the lack of data, but in the difficulty of converting them into integrated information that can be interpreted and used in the supervision of public procurement. In analysing the records only through tables, it is complex to determine which entities repeatedly award contracts to the same suppliers, which suppliers are shared by different institutions, which actors maintain a prominent participation and in which cases there is a greater concentration of awards by frequency or amount.

According to Burgos Carlosama (2025), the network analysis applied to public procurement allows to study the interactions between buyers and suppliers and to identify relationships and relevant actors that are not easily distinguished by an exclusively tabular analysis. This approach makes it possible to understand that each award is not only an isolated record, but also part of a connection structure generated by the repeated participation of entities and suppliers in different procedures.

The difficulty in visualizing these connections limits the possibility of understanding how the market for the Electronic Reverse Auction is organized and of prioritizing processes that may require a more detailed review. In addition, Redroban Ortiz et al. (2024) point out that the analysis of complex networks allows to examine the connectivity of the participants and to recognize the actors that occupy

relevant positions within public procurement. However, a recurrent relationship, a high participation or a concentration of awards do not in themselves constitute evidence of irregularity. These results should be interpreted as analytical signals to identify patterns and guide subsequent monitoring procedures.

To respond to this problem, the research proposes to build in Neo4j a two-party network of the procedures of Electronic Reverse Auction initiated in Ecuador between January and December 2025. A two-party network is made up of two different groups of nodes: the first represents the contracting entities and the second represents the awarded suppliers. The awards are the relationships that connect both groups. By their structure, the connections are established between an entity and a supplier, and not between actors belonging to the same group.

This representation will allow the consolidation of the awards of each entity-supplier relationship and a joint analysis of its frequency and amount. It will also facilitate the identification of recurrent links, shared suppliers, high-participation actors, communities, subnetworks and possible concentration patterns. In this way, Neo4j will allow to transform the dispersed records of SERCOP into a relational structure that facilitates its exploration and generates clearer information to support monitoring, prioritization of reviews and public decision-making.

Based on this problem, the research seeks to answer the following question: How can a bipartite network model developed in Neo4j facilitate the analysis of relations, participation, recurrence and concentration between the contracting entities and the suppliers awarded in the procedures of Electronic Reverse Auction in Ecuador initiated in 2025?

5. General Objective

Develop a network analysis model in Neo4j to represent and analyse the relationships between contracting entities and suppliers in the procedures of Electronic Reverse Auction in Ecuador initiated in 2025, in order to identify concentration patterns, recurrent links and relevant actors that provide useful information for monitoring and decision-making.

6 Specific Objectives

- Collect, integrate, and clean SERCOP open data for Electronic Reverse Auction procedures initiated in 2025, ensuring consistency across entities, suppliers, awards, and amounts.
- Build a graph model in Neo4j that clearly represents the connections among contracting entities, suppliers, and awarded procedures.
- Analyze the frequency and economic value of awards to identify suppliers with high participation, recurrent relationships, and possible concentration levels.
- Apply network metrics to identify relevant central actors, communities, and structures within the market analyzed.
- Generate visualizations and findings applicable to public management that facilitate data understanding and guide monitoring and review prioritization, without treating detected patterns as direct evidence of irregularity.

7. Justification And Application Of Methodology

7.1. Data collection

The data set was obtained from the National Public Procurement Service (SERCOP) open data portal. Because it corresponds to administrative records previously published by a State institution, it was used as an official secondary source. The study included the procedures of Electronic Reverse Auction initiated between 1 January and 31 December 2025.

The information was manually downloaded by filters per year, month and modality. The consultation was conducted separately for each month and produced 12 JSON files structured according to the Open Contracting Data Standard (OCDS). At this stage, all

available records were kept, without any exclusions or modifications. The original files were stored in the data/raw/2025 directory and incorporated into the project repository to ensure their traceability and allow their subsequent processing in Python and Neo4j.

Table 2 Quantitative summary of the data collected

Item	Quantity or characteristic	Description
Releases collected	18,326	Structured publications containing procedure information.
Distinct OCIDs	18,326	Unique identifiers for the procedures collected.
Contracting entities	1,737	Preliminary total before verifying which entities recorded awards.
Bidder participations	79,425	Participation records; one bidder may appear multiple times.
Distinct bidders	12,125	Unique actors initially identified as participants.
Procedures with an award	15,718 (85.77%)	Procedures containing award information.

Note. Own preparation based on SERCOP open data (2026). OCDS = Open Contracting Data Standard; OCID = Open Contracting ID.

7.2. Cleaning, pre-processing and / or processing of data.

Once the monthly files were integrated, the cleaning and preparation of the procedures, participation of bidders and awards were carried out in Python. The OCID identifiers and the procedural combinations - offering and procedure - were verified and the supplier was not found to be duplicated. It was also confirmed that buyer.id and tender.procuringEntity.id identified the same contracting entity.

To consistently recognize buyers, bidders and suppliers, the identifier was used as a main reference and the name as a descriptive attribute. The original names were retained and standardized versions were created to reduce differences in capital letters, tides and spaces. The missing names were recovered only when the same identifier existed in other records. Atypical identifiers and consortia were maintained as published by the SERCOP and were marked by quality control variables.

In the economic variables, the reference value was recovered mainly from tender.lots, and four outstanding assigned amounts were obtained from correctedValue. After treatment, no amounts were awarded, negative or zero. The 293 procedures without information on the number of bidders and the fields without sufficient support to be completed were kept as null and void, without artificial charges.

The temporary validation confirmed that all procedures began in 2025. 1,858 awards registered in 2026 were retained because

they were in the case of procedures initiated during the period under review. To study the recurrence cpc _ 5 was created from the CPC code. The 1,831 extreme amounts identified were also retained, because a high value does not necessarily represent an error. The log10 transformations and the 99 percentile winsorization were used only for visualizations,

without changing the amounts used in the analysis. The three final bases retained the original identifiers, traceability variables and quality flags. The details of the validations and treatments applied are presented in Table A1 of Annex 1.

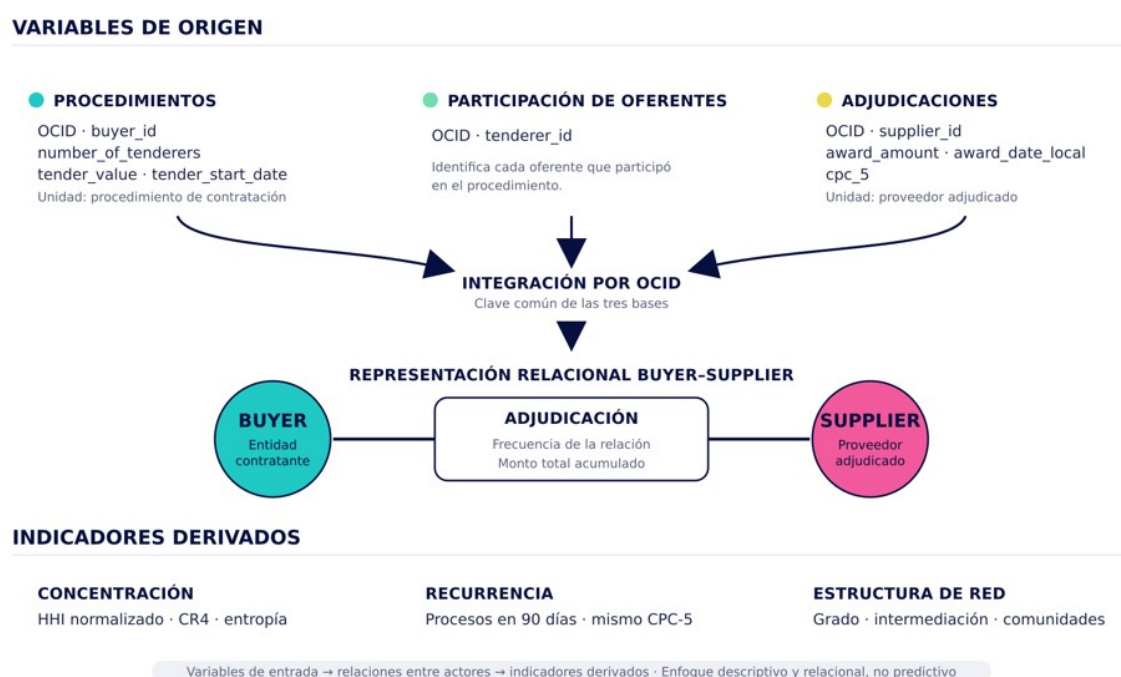
7.3. Identification and description of variables

After cleaning and pre-processing, open data published by the National Public Procurement Service (SERCOP, 2026) were organized on three related bases through OCID: procedures, with 18,326 records; participation of bidders, with 79,425 records; and awards, with 15,718 records. Each base retains a different analysis unit, which allows the procedures, participating bidders and awarded suppliers to be studied without mixing observations of a different nature.

The variables were classified according to their function. The oid, buyer _ id, tenderer _ id and supplier _ id identifiers allow for the recognition of procedures and actors. The number of bidders, the reference value, the amount awarded, the dates and the cpc _ 5 code function as input variables. From these, the indicators of concentration, recurrence and network structure, including standardized HHI, CR4, entropy, degree and betweenness centrality, were calculated. Because the study is descriptive and relational, no objective or predictive variables were defined in the statistical sense. The dictionary of the variables is complete in Table A2 of Annex 2.

Figure 1 summarizes how the three datasets analyzed are organized and integrated through the OCID. It shows the input variables, the links between contracting entities and awarded suppliers, and the derived indicators. This structure represents a descriptive, relational, and non-predictive approach.

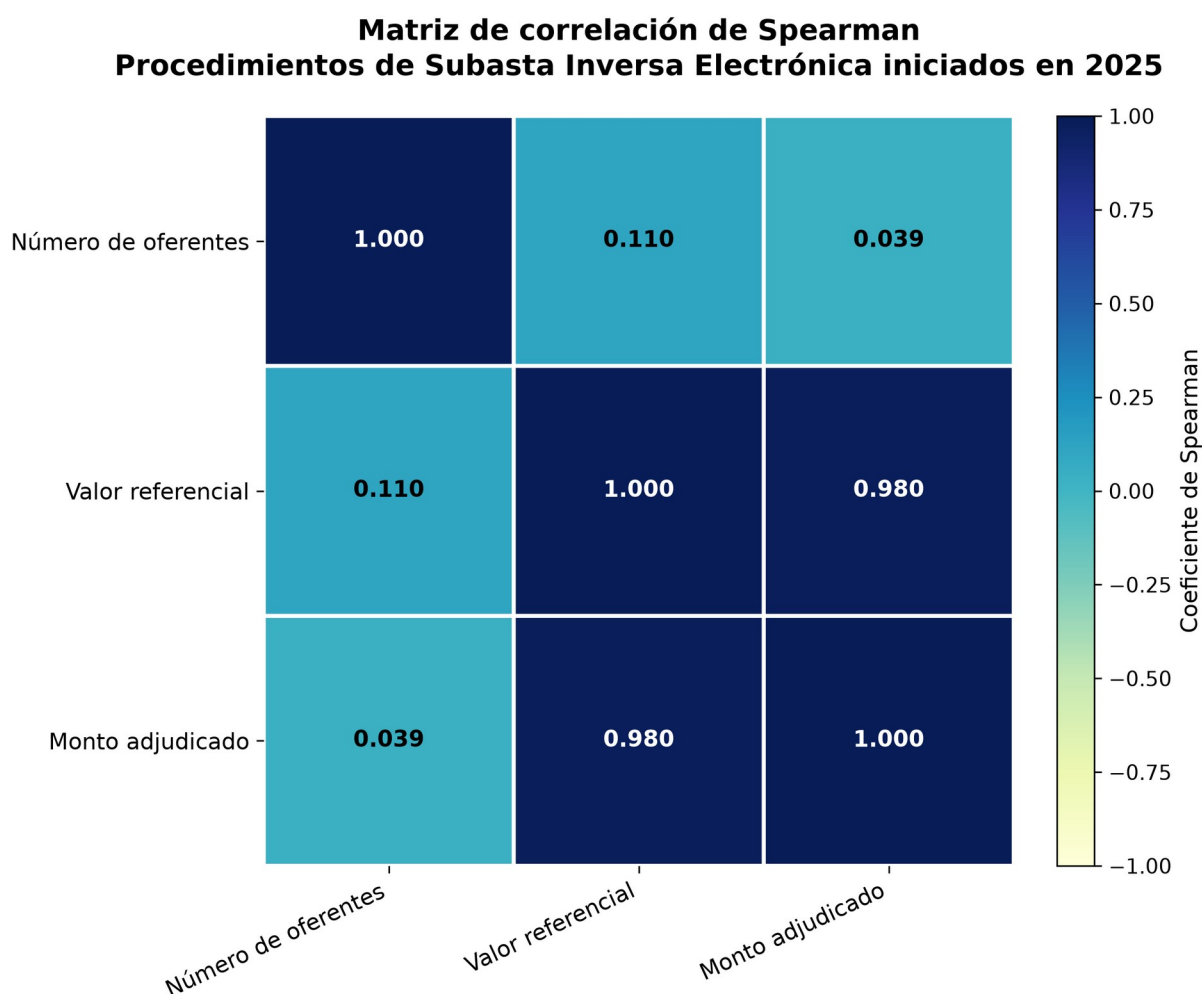
Figure 1 Available variables and their relationships



Note. Own production based on SERCOP open data for 2025.

Figure 2 presents the Spearman correlations calculated for 15,714 procedures. A very strong association was found between the reference value and the awarded amount ($\rho = 0.980$), while the relationships between bidder count and the monetary variables were weak or null ($\rho = 0.110$ and $\rho = 0.039$). Awarded amount therefore does not by itself explain participation. The analysis was supplemented with concentration, recurrence, and network indicators; these associations do not imply causation.

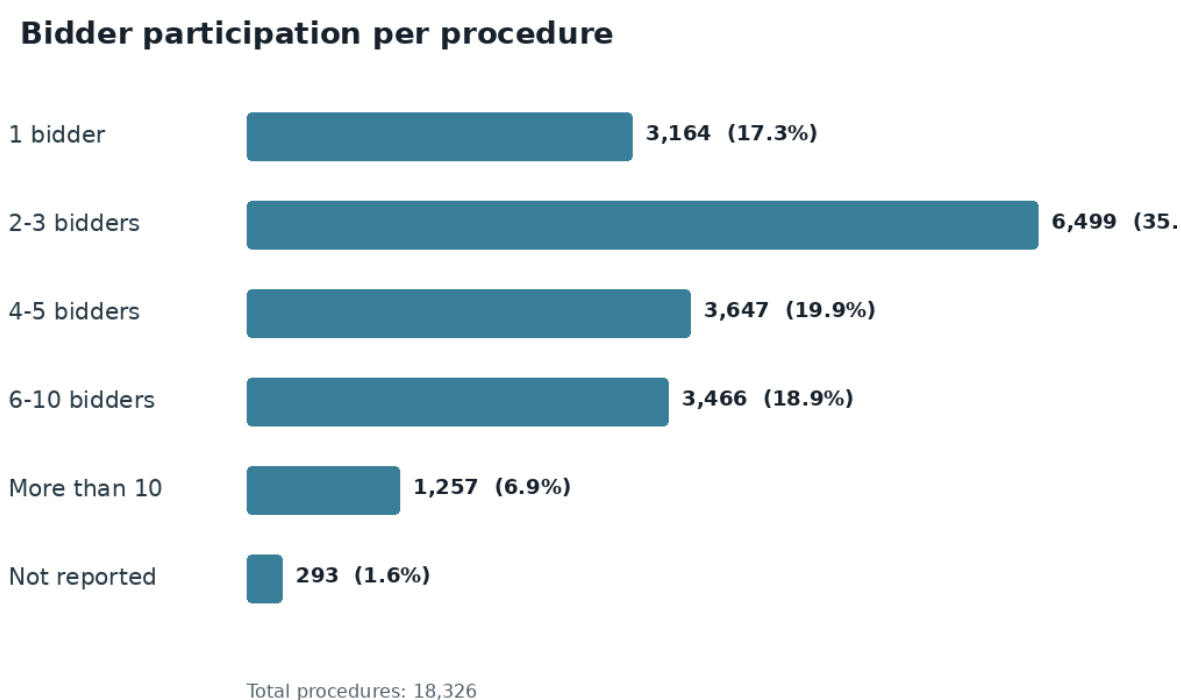
Figure 2 *Spearman correlation matrix for the numerical variables analyzed*



Note. Own production based on SERCOP open data for 2025.

7.4 Variable Visualization

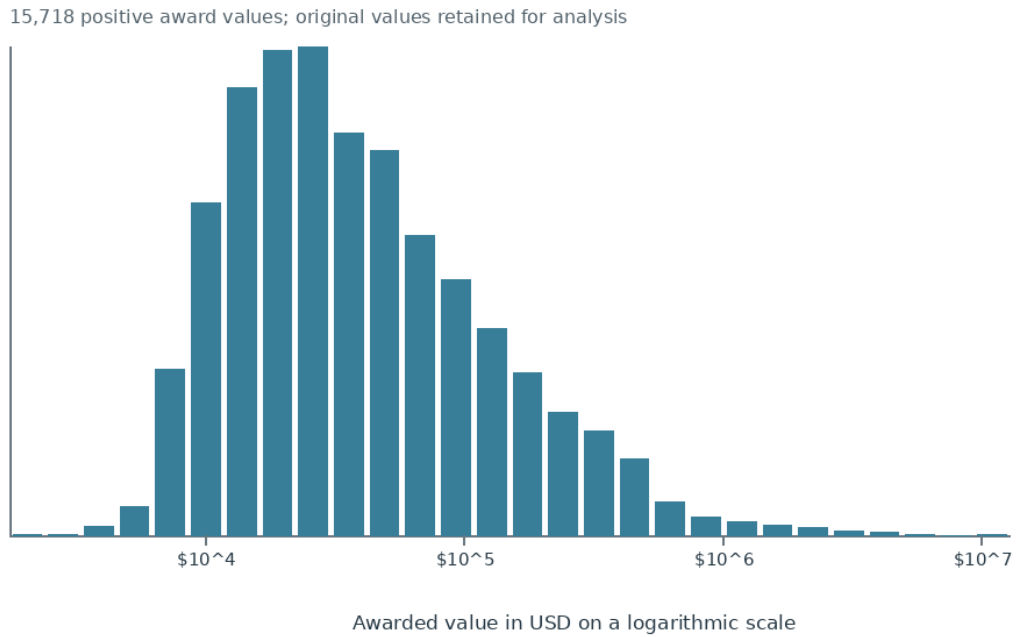
To examine the behavior of the variables, bidder participation and the distribution of awarded amounts were analyzed. These visualizations show competition levels and the economic dispersion of procedures before application of the network indicators.

Figure 3 Bidder participation per procedure

Note. Own production based on SERCOP open data for 2025.

Figure 3 shows that 52.8% of procedures had no more than three bidders: 17.3% had one bidder and 35.5% had two or three. Only 6.9% had more than ten bidders, and 1.6% did not report this variable.

Figure 4 shows a strongly right-skewed distribution. The median was USD 32,601.26, the 99th percentile was USD 1,006,458.20, and the maximum was USD 12,840,063.97. A base-10 logarithmic scale was used for visualization without removing records or modifying the original amounts used in calculations.

Figure 4 Distribution of awarded amounts**Distribution of awarded values**

Note. Own production based on SERCOP open data for 2025.

Together, Figures 3 and 4 show participation concentrated among a small number of bidders and wide dispersion in awarded amounts. Consistent with Burgos Carlosama (2025), these dimensions should be analyzed separately and treated as contextual signals rather than evidence of irregularity.

Additional visualizations were developed for suppliers' economic participation, award counts, and concentration by amount and frequency. Because they require greater detail, these figures are presented in Annex 3 as support for the analytical results.

7.5 Statistical Model Selection

Because the study describes and analyzes relationships between contracting entities and suppliers, a descriptive network-analytics model rather than a predictive model was selected. Newman (2003) and Sayama (2015) explain that complex systems can be examined by representing actors as nodes and their interactions as connections. The data were therefore modeled in Neo4j as a bipartite Buyer-Supplier network in which buyers and suppliers form two distinct node sets connected exclusively by recorded awards.

$$G = (V, E)$$

$$V = B \cup S$$

Here, G represents the network, V the complete node set, and E the award relationships. B represents contracting entities and S awarded suppliers. Each `CONTRACTS_WITH` relationship connects an entity and supplier and consolidates the awards between them.

7.5.1 Relationship Properties

Two principal properties were calculated for each Buyer-Supplier relationship: frequency and accumulated amount. Frequency is the number of distinct procedures awarded between contracting entity b and supplier s .

$$f_{b\ s} = n_{b\ s}$$

Accumulated amount is the sum of the values awarded across all awards associated with the same relationship.

$$m_{b\ s} = \sum_{k=1}^{n_{bs}} M_{b\ s\ k}$$

These properties separate repetition of a relationship from its economic importance. Consistent with Burgos Carlosama (2025), frequency and economic weight were analyzed jointly but distinguished. For visualization, amount was transformed using a base-10 logarithm to reduce the visual influence of extreme values; original amounts were retained for calculations.

7.5.2 Concentration of Awards

Concentration was evaluated separately by awarded amount and award frequency because these dimensions represent different behaviors. The Herfindahl-Hirschman Index (HHI) sums the squared shares of each supplier within an entity's awards (U.S. Department of Justice and Federal Trade Commission, 2010).

$$HHI_b = \sum_s p_{b\ s}^2$$

In this expression, p_{bs} is supplier s 's share of contracting entity b 's awards, calculated by amount or frequency. To compare entities with different numbers of suppliers, the normalized HHI was used.

$$HHI_b^* = \frac{HHI_b - \frac{1}{n_b}}{1 - \frac{1}{n_b}}$$

7.5.3 Recurrence and Relationship Prioritization

Recurrence was not defined solely by the total number of awards. It was defined as repetition of the same contracting entity, supplier, and contractual object identified by the CPC-5 code within a 90-day window.

$$R_{b s c} = 1 \text{ si. } N_{b s c, 90} > 3$$

In this expression, b represents the contracting entity, s the supplier, c the CPC-5 code, and $N_{bsc,90}$ the maximum number of distinct procedures found for that combination within 90 days. A relationship was classified as recurrent when it recorded more than three procedures, meaning at least four.

Prioritization combined concentration, recurrence, single-bidder participation, and awarded-amount signals. Thresholds were justified through percentiles, minimum observation counts, and sensitivity analysis using alternative cutoffs. Consistent with Redroban Ortiz et al. (2024), these relational metrics support control and oversight. The combined signals rank relationships requiring review; they do not establish the existence of irregularities.

7.5.4.

Bipartite Network

For structural analysis, the bipartite network was projected as not directed at Neo4j GraphData Science, with the name buyer _ supplier _ 2025. Weakly Connected Components (WCC) was applied to identify connected components and the degree to measure the direct connections of each actor. In addition, the betweenness centrality proposed by Freeman (1978) was calculated to recognize entities or suppliers that function as bridges between different parts of the network. The latter was estimated by a reproducible sample of 1,000 nodes and a seed of 42.

Community detection was carried out with Louvain, an algorithm developed by Blondel et al. (2008) to group nodes by optimizing modularity. As a complementary method, Leiden was applied, proposed by Traag et al. (2019), which favours internally connected communities. The results were compared by the number of communities, modularity and distribution of their sizes; in addition, the convergence of the algorithm was verified in Leiden.

The Cypher consultations, the construction of the projection, parameters, validations and complete outputs are found in the network analysis notebook available in the GitHub repository of the project (see Annex 4). The joint application of these algorithms made it

possible to analyse the organization of the network, its subnetworks, communities and brokering positions.

8. Results

8.1. Model analysis

This section presents the results obtained by applying the Buyer-Supplier bipartite network model built in Neo4j. For structural analysis, the Neo4j GraphData Science (GDS) library was used to examine network connectivity, identify central actors and brokering positions, identify communities and analyse shared suppliers. Concentration, recurrence and signal combination were also evaluated. The graphs synthesize the main findings, which allow prioritizing relationships for a later review, but do not alone demonstrate the existence of irregularities.

8.1.1.

Network structure and connectivity

The network consolidated 15,718 awards in 13,493 Buyer-Supplier relations, because the same entity was able to award more than one procedure to the same supplier. The projection was made up of 7,664 actors: 1,695 contracting entities and 5,969 suppliers awarded.

Table 3 Structural summary of the Buyer-Supplier network

Indicator	Result
Contracting entities with awards	1,695
Awarded suppliers	5,969
Total actors	7,664
Buyer-Supplier relationships	13,493
Weakly connected components	222
Largest component	7,097 actors (92.6%)

Note. Actors are contracting entities and suppliers with at least one award. Own preparation based on Neo4j results using SERCOP open data for 2025.

The analysis using weakly connected components showed that 7,097 actors, equivalent to 92.6% of the network, belong to the same main component. The remaining actors were distributed into 221 smaller components, the median of which was two. This shows a widely connected structure, along with isolated subnetworks that can respond to specialization, territory or less contractual activity.

Figure 5 Selected Buyer-Supplier Network Supports

Note. Green nodes represent contracting entities and pink nodes, awarded suppliers. Each connection corresponds to a consolidated Buyer-Supplier ratio. Own production based on results obtained in Neo4j with open SERCOP data for 2025.

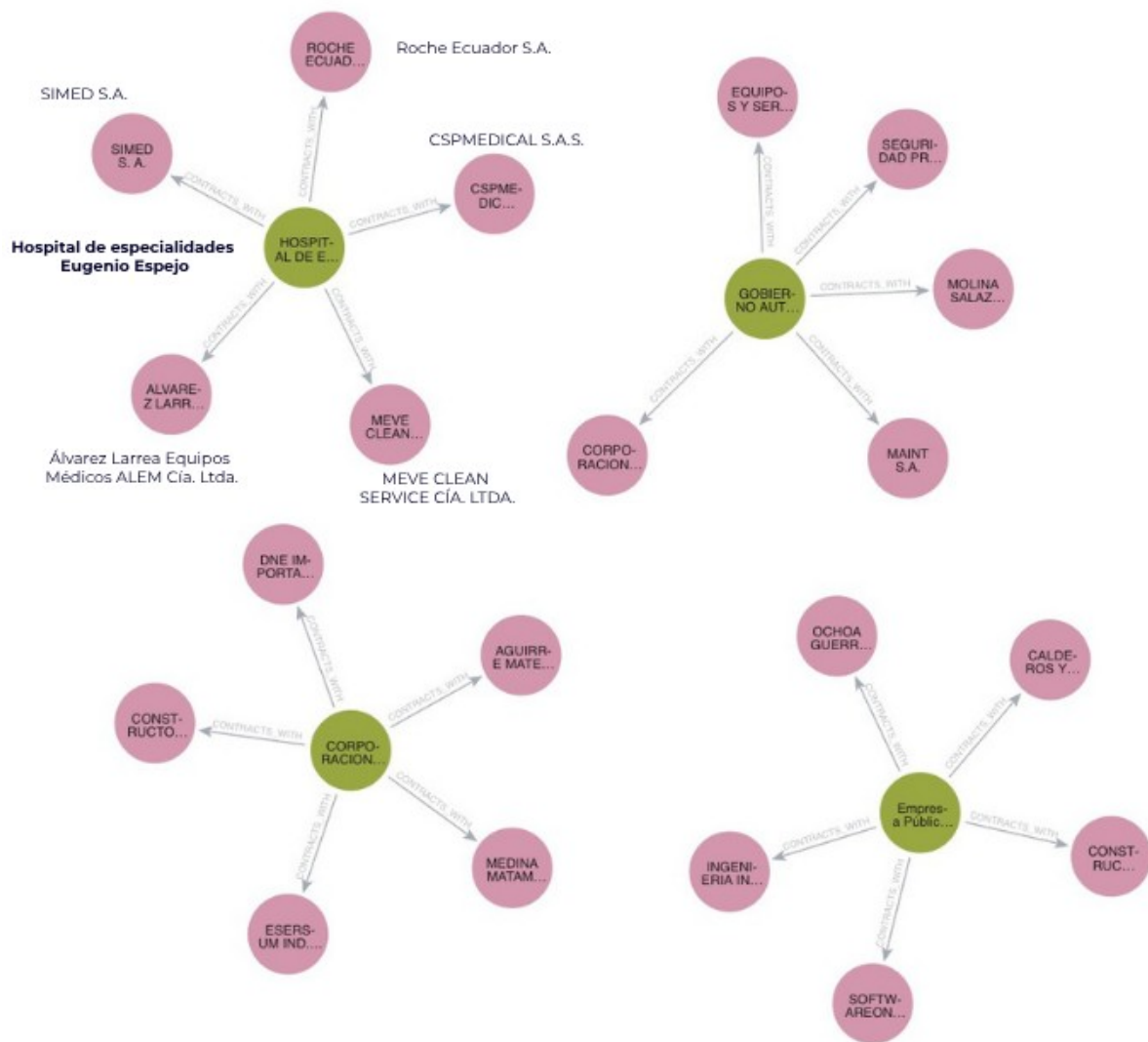


Figure 5 presents a partial map of the Buyer-Supplier network. Four subnetworks were selected to preserve legibility, as showing the 7,664 actors simultaneously would make it difficult to interpret them. As an example, the Eugenio Espejo Speciality Hospital is connected with suppliers such as SIMED S.A., Roche Ecuador S.A., Álvarez Larrea Teams Médicos, CSPME and MEVE Clean Service, among others. Each connection represents a consolidated award relationship between the entity and the supplier. The figure allows for the recognition of related buyers and suppliers, but it represents only a part of the network and does not evidence irregularities.

8.1.2.

Central actors and brokering

The metrics identified different forms of centrality. CELEC EP presented the greatest diversity of relationships, connecting with 312 suppliers, and also recorded the highest frequency, with 457 awards. In contrast, EP Petroequator concentrated the largest amount accumulated, with USD 54.271.908.42. These results show that the degree, frequency and amount describe different dimensions of the actors' participation.

Figure 6 Actors with greater betweenness centrality

Note. The betweenness centrality was estimated by a reproducible sample of 1,000 nodes. Own production based on results obtained in Neo4j with open SERCOP data for 2025.

Figure 6 presents a different metric: the betweenness centrality. In this case, the GAD of the Metropolitan District of Quito held the first position, with 542,576.14, and CELEC EP the second, with 466,448.48. This means that both actors often appear on routes that connect different parts of the network. CELEC therefore leads in diversity and frequency, but it is second in betweenness centrality. These positions describe their structural function and do not show irregularities.

8.1.3.

Communities identified on the network

To analyze the internal organisation of the network, the Louvain algorithms (Blondel et al., 2008) and Leiden (Traag et al., 2019) were compared. Louvain identified 260 communities with a modularity of 0.6872, while Leiden found 263 communities with a modularity of 0.6860 and confirmed the convergence of the algorithm.

Table 4 Comparison of identified communities

Algorithm	Communities	Modularity	Largest community
Louvain	260	0.6872	1,033 actors
Leiden	263	0.6860	1,063 actors

Note. Own preparation based on Neo4j results using SERCOP open data for 2025.

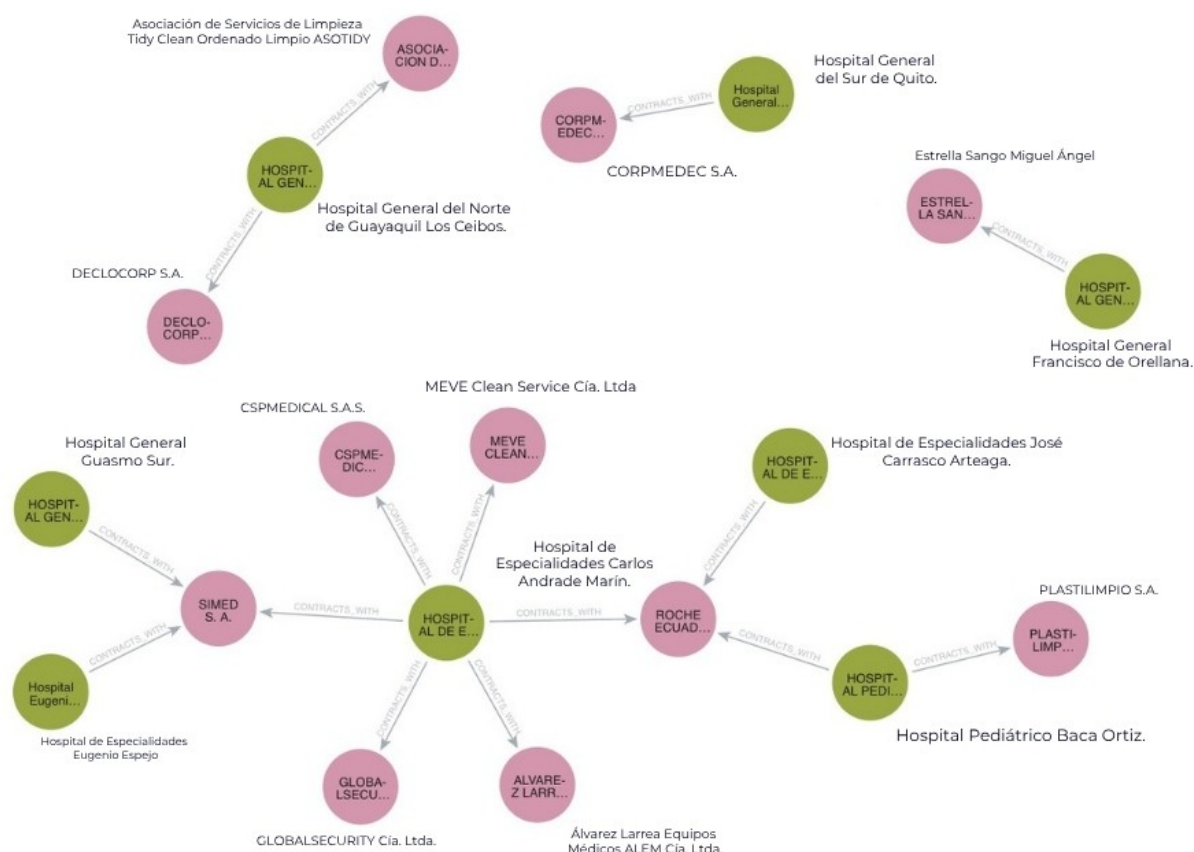
The results of both algorithms were similar. A modularity of about 0.69 indicates that connections are more frequent within communities than between different communities. Leiden was taken as a supplementary reference because it confirmed convergence and encouraged the formation of internally connected groups.

Figure 7 Fragment of a community detected by Leiden

Note. Green nodes represent contracting entities and pink ones, awarded suppliers. There is a fragment of the community and not its 1,063 members. Own production based on results obtained in Neo4j with open SERCOP data for 2025.

Figure 7 shows a fragment of a community detected by Leiden. The hospitals are Carlos Andrade Marín, Baca Ortiz, José Carrasco Arteaga, Guasmo Sur and Eugenio Espejo, connected by shared suppliers such as Roche Ecuador S.A.. The representation by nodes allows for the rapid recognition of this grouping.

In a table, these relationships would appear in separate rows and the pattern would be less evident. According to Lyra et al. (2021), community analysis allows for the identification of structures that are not observed when studying the procedures individually. The CPC-5 allows to check whether the connections correspond to similar purchases or to different contractual objects. Therefore, the network identifies



the pattern and the CPC-5 provides the context to interpret it, without demonstrating irregularities.

8.1.4.

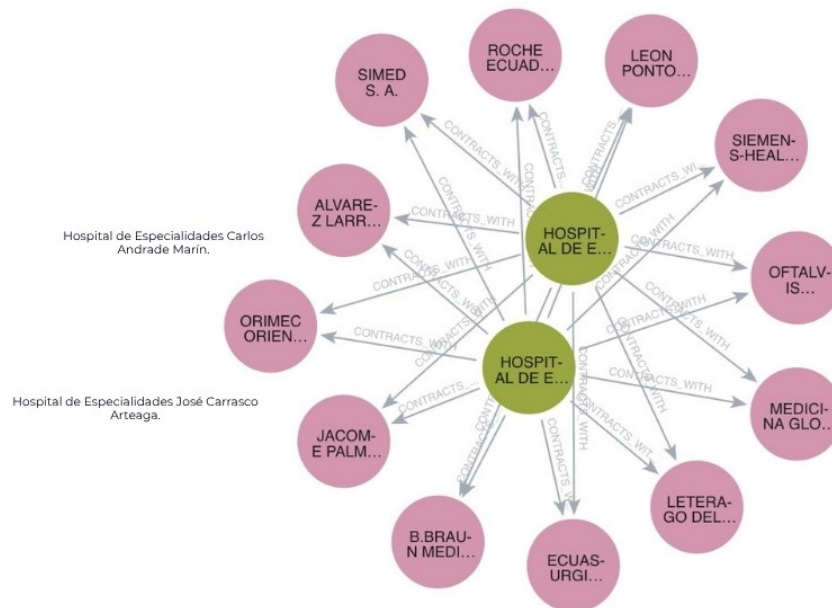
Reason for shared suppliers

The analysis of the reasons for this made it possible to identify related contracting entities through common awarded suppliers.

Figure 8 Structural reasons for suppliers shared between two contracting entities

Note. Green nodes represent contracting entities and pink ones, awarded suppliers. Own production based on results obtained in Neo4j with open SERCOP data for 2025.

Figure 8 shows that the Carlos Andrade Marín Speciality Hospital and the José Carrasco Arteaga Speciality Hospital shared 54 suppliers. The display has only the 12 suppliers with the largest combined amount. This pattern may be related to similar recruitment needs, but it does not demonstrate coordination or irregularities.



8.1.5.

Concentration of awards

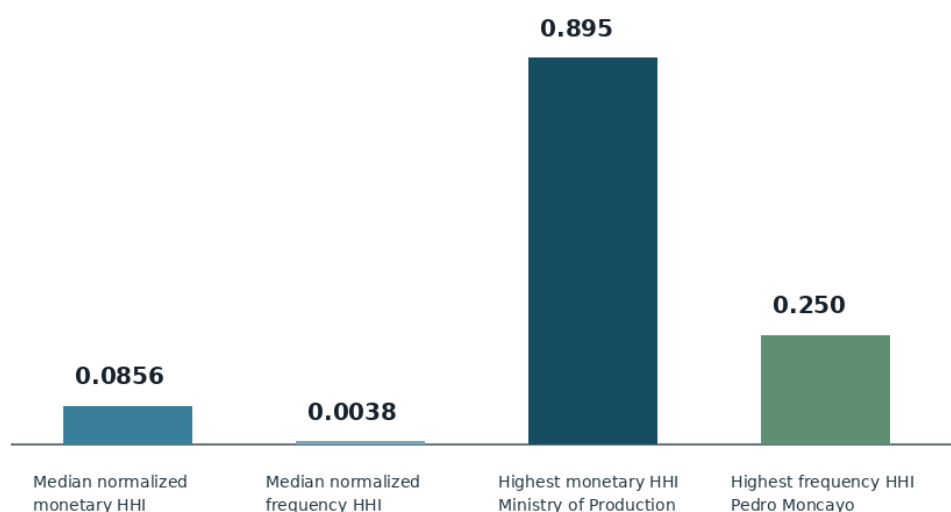
Figure 9 Standardized concentration of awards by amount and frequency

Note. The standard HHI was calculated for 687 buyers with at least five awards, considering amounts equal to or greater than USD 5,000. Own production based on results obtained in Neo4j with open SERCOP data for 2025.

The median of the standardized monetary HHI was 0.0856, while the median by frequency was 0.0038. This shows that an entity can distribute its awards to several suppliers while at the same time concentrating a higher proportion of the amount on one of them. For this reason, both indicators were interpreted separately.

Concentration differs by monetary value and frequency

Normalized HHI ranges from 0 (dispersed) to 1 (concentrated).



8.1.6.

Recurrence of relationships

Figure 10: Sensitivity of the recurrence criterion

Note. Own production based on results obtained in Neo4j with open SERCOP data for 2025.

Recurrence considers the same combination of contracting entity, supplier and CPC-5 within a 90-day window. 111 relationships were identified with more than two procedures, 23 with more than three, 7 with more than four and none with more than five. As a final rule, the threshold of more than three procedures was selected, i.e. a minimum of four for the same Buyer-Supplier and CPC-5 combination within 90 days.

8.1.7.

Final prioritization of relations

Sensitivity of the 90-day recurrence criterion

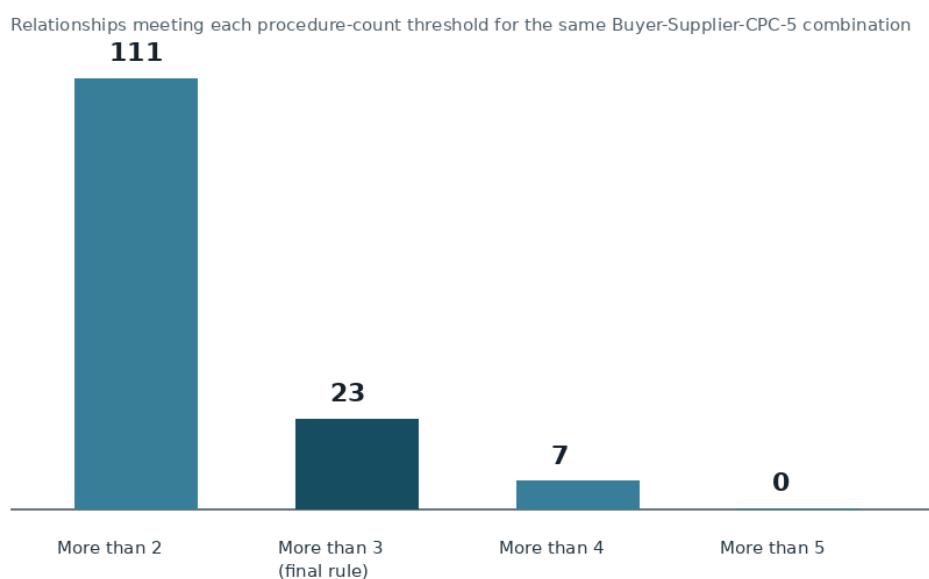


Figure 11 Prioritization of relationships by the combination of signals

Note. Own production based on results obtained in Neo4j with open SERCOP data for 2025.

Prioritization incorporates minimum observation controls to prevent percentages calculated on few procedures from generating artificial signals. Of the 13,493 relationships evaluated, 71 gathered three or more signals, equivalent to 0.53%, and 3 presented four signals, equivalent to 0.02%. The control of small denominators reduced the relationships with three or more signals from 257 to 71 and those with four from 10 to 3. The result makes it possible to set priorities for a further review, but it does not confirm irregularities.



9. Discussion Of Results And Proposal For Solution

9.1. Interpretation of results

The analysis made it possible to observe public procurement as a network of relationships and not as isolated procedures. This perspective facilitated the recognition of connections between entities and suppliers that would be difficult to identify through conventional tables.

92.6% of the actors were concentrated in a major component. This shows that most of the network is connected directly or indirectly, while small groups can respond to specialized purchases, territorial differences or less contractual activity.

The relevant actors changed according to the metric used. CELEC EP highlighted for its number of suppliers and awards; EP Petroequator, for the amount accumulated; and the Quito GAD, for its betweenness centrality. Burgos Carlosama (2025) argues that the frequency and amount should be analysed together, but separately. Therefore, no individual metric shows an irregular situation.

Louvain and Leiden identified a similar community structure, with a modularity of about 0.69. This means that the actors in a community are more related to each other than to the rest of the network. In one of the fragments, several hospitals were observed connected by common suppliers. In addition, the hospitals Carlos Andrade Marín and José Carrasco Arteaga shared 54 suppliers.

This finding demonstrates the usefulness of visualizing the information as a network. According to Lyra et al. (2021), communities allow for the recognition of patterns that are not evident when studying each procedure individually. However, it is

need to review the CPC-5 and the contractual object to determine whether the connections correspond to similar purchases.

The monetary concentration was higher than the concentration per frequency. The Ministry of Production presented the highest monetary HHI (0.895), while Pedro Moncayo led the concentration by frequency (0.250). This difference confirms that the concentration of a high proportion of the amount does not amount to the repeated award of the same suppliers. Burgos Carlosama (2025) recommends analyzing both dimensions separately, and Pliatsidis (2025) points out that their interpretation should consider the sector, the contractual object and the availability of suppliers.

Recurrence also changed according to the applied threshold. The final rule identified 23 relationships with more than three procedures for the same buyer, supplier and CPC-5 within 90 days. The minimum of observations prevented percentages calculated with few procedures from being interpreted as sustained patterns.

Finally, the signal combination reduced the 13,493 relationships to 71 with three or more signals and to 3 with four. This result makes it possible to direct the review to a manageable group and to know the reasons for its selection. As Redroban Ortiz et al. (2024) points out, network analysis can guide monitoring, but the determination of an irregularity requires a review of the files and context of each case.

9.2. Implications for the organization

For SERCOP, the main contribution of the model is to convert the available data into a monitoring priority mechanism. The Organic Law of the National Public Procurement System (LOSNCP, 2025, arts. 9, 13 and 21) provides for the use of open data, technological tools and objective criteria, as well as early warning systems to identify patterns such as repeated concentration of awards. In this context, the 71 relationships with three or more signals could form a first review group, while the 3 relationships with four signals would have the highest priority.

Network analysis would also allow for a shift from the isolated review of procedures to the joint analysis of entities, suppliers and sectors. For example, hospital communities and shared suppliers could be monitored by CPC-5, dates, number of bidders, amounts and technical specifications. This would help to differentiate relationships with normal sector needs from those with recurrence, concentration or low participation of bidders and require more detailed examination.

Prioritization should be complemented by a technical and documentary review. The OECD (2026) warns that indicators can respond to market structure, institutional conditions or integrity risks; they should therefore motivate further research and not be interpreted as evidence of an irregularity. The model would therefore guide the review of files and,

where appropriate, facilitate coordination with the competent control bodies, in accordance with Article 14 of the LOSNCP.

In the area of competition, SERCOP could identify the CPC-5 categories that combine low participation of bidders and high concentration by amount or frequency. This would provide guidance on market studies and assistance to contracting entities prior to new procedures. As the OECD points out (2026), low competition can respond to few suppliers available, geographical restrictions or process conditions, so each market should be analysed according to its context.

Another involvement corresponds to the quality of the data. The missing or inconsistent fields of bidders, suppliers and CPC-5 codes reduce the accuracy of indicators and alerts. Therefore, the SERCOP could incorporate automatic validations when recording the information and generate quality reports for the contracting entities. This measure would strengthen the monitoring and production of reliable statistics, in accordance with articles 9 and 21 of the LOSNCP (2025).

Finally, the analysis could be carried out on a regular basis to compare results between years, to assess signal stability and to adjust the thresholds by means of expert review. Because the developed model is descriptive and non-predictive, it would function as a first analytical layer to support the future development of more complete institutional alerts.

10. Conclusions And Recommendations

10.1. Conclusions

The main finding was that the public procurement analysed does not work as a set of isolated procedures, but as a relationship structure. The network formed by 13,493 Buyer-Supplier links allowed the recognition of subnetworks, central actors and connections that were not visible on the stool bases.

Within this structure, the relevance of the actors was multidimensional. Having more suppliers, receiving more awards, accumulating a high amount or holding a brokering position represented different functions within the network. Therefore, no individual ranking was sufficient to explain the importance of an entity or supplier.

A sectoral pattern was also identified in the community selected by Leiden: the observed contracting entities were mainly hospitals linked by common suppliers. The fact that the hospitals Carlos Andrade Marín and José Carrasco Arteaga shared 54 suppliers demonstrated the existence of a related supply environment, the nature of which must be specified by CPC-5 and the contractual object.

Competition and concentration also did not have a unique behaviour. 52.8 per cent of the procedures recorded no more than three bidders, while the concentration by frequency and amount produced different results. This confirmed that a limited participation or a high HHI should be analysed together with other characteristics of the procedure.

The central conclusion of the project was that the combination of signals provides more analytical capacity than any isolated indicator. Of the 13,493 relationships studied, 71 gathered three or more signs and only three presented four. The model thus managed to transform an extensive network into a

reduced and justifiable universe of priority cases, without considering the signs as evidence of irregularity.

Overall, the network analysis complemented the traditional analysis of public procurement by incorporating structure, context and recurrence. The model is reproducible for other periods, but its results depend on the quality of the data and require documentary and expert validation before issuing conclusions on a specific relationship.

10.2. Recommendations

It is recommended to use the 71 relationships with three or more signals as a first monitoring group and to consider the 3 relationships with four signals as cases of higher priority. This classification should guide the use of institutional resources and not function as an automatic sanction.

Each priority relationship should be validated by reviewing the file, the CPC-5, the participants, the dates, the amounts, the technical specifications and the justification for the award. As the OECD notes (2026), risk indicators require further research to consider the characteristics of the market and the sector analysed.

It is also recommended to run the model regularly and to incorporate it into a monitoring board that compares concentration, recurrence, participation of bidders and betweenness centrality. The results of the expert review would make it possible to adjust the thresholds, reduce false positive and assess the stability of signals between different years and procurement modalities.

Communities and subnetworks could be used to conduct sector-focused reviews. In the case of hospitals, the CPC-5 analysis would allow for the recognition of shared suppliers, dependence on certain actors and possible supply alternatives, as well as the comparison of purchasing conditions between entities with similar needs.

As a future research, it is recommended to expand the Neo4j network through data on inventories, delivery times, non-compliance, supply capacity and availability of suppliers. The Pan American Health Organization (PAHO, 2024) recognizes the management of the supply of medicines and health technologies as a necessary function of health services. With these variables, network analysis could help to identify critical suppliers, possible bottlenecks and supply alternatives in the face of interruptions affecting hospital entities.

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12. Annexes

Annex 1 Data Cleaning and Preprocessing Validations

Table A1 Identified data issues validations and treatments

Identified issue	Validation result	Treatment applied
Duplicate OCIDs	0	Verified that each procedure had one unique OCID. No records were removed as duplicates.
numberOfTenderers not reported	293 procedures	Retained as missing data; not replaced with zero because absence of information does not establish that a procedure had no bidders.
Differences between numberOfTenderers and registered tenderers	0 differences in 18,033 comparable cases	Compared SERCOP's reported count with the number of registered bidders. No correction was needed.
Bidder names recorded as null	27 records	Searched the same tenderer_id elsewhere. Recovered 20 names using identifier evidence; 7 remained unnamed because information was insufficient.
Supplier names recorded as null	9 records	Searched the same supplier_id in other awards and parties. Recovered 4 names; 5 remained unnamed without removing the records.
Variations in entity and supplier names	Same ID associated with different spellings	Retained the original SERCOP name and created a normalized comparison version removing differences in case, accents, and spaces. The identifier remained the principal reference.
Atypical buyer identifier	1 procedure	Retained the identifier exactly as published and created identifier_comprador_atipico as a control variable.
Atypical bidder identifiers	52 participations involving 12 bidders	Checked against parties and other records. With no supported alternative version, identifiers were retained and marked with a quality-control variable.
Atypical supplier identifiers	23 awards	Retained the original identifiers and created a quality flag. No digits were removed and no tax IDs were constructed manually.
Same tax ID associated with multiple identifiers	0 cases	Verified that one supplier was not fragmented across different IDs. No tax-ID mergers were needed.
Consortia	8 consortia in 24 awards	Retained as independent suppliers under their own identifiers. Awards were not distributed among member firms because composition information was insufficient.
Missing tender.value	18,326 procedures	Obtained the reference value from tender.lots[0].value, where SERCOP published the amount for the procedures analyzed.
Procedure without lot information	1 procedure	Retained for traceability and flagged with registro_atipico_calidad. No artificial reference value was assigned.
Missing award.value	4 awards	Used correctedValue, which contained the amount published by SERCOP, and created a variable identifying the source field.
award.status not reported	15,718 awards	Retained as null; status was not

Identified issue	Validation result	Treatment applied
		inferred from other procedure fields.
tenderPeriod.endDate not reported	18,326 procedures	Retained as null; not substituted with awardPeriod, enquiryPeriod, or another date because they represent different stages.
Procedures without an award	2,608	Retained in the general procedures table; excluded only from analyses requiring an awarded supplier or amount.
Procurement method	18,326 Electronic Reverse Auction records	Verified that the entire dataset corresponded to this method. No Electronic Catalogue records were present.
Dates apparently outside 2025 after UTC conversion	27 cases initially shifted to 2026	Used the local date published by SERCOP to define the procedure year; all 18,326 procedures began in 2025 under this criterion.
Awards recorded in 2026	1,858	Retained because the procedures began in 2025 and were awarded later; these were not treated as new 2026 procedures.
Awards predating procedure start	0	Compared award and procedure start dates; no temporal inconsistencies were found.
Differences between buyer.id and tender.procuringEntity.id	0	Confirmed that both fields identified the same entity in all 18,326 procedures.
Currencies other than USD	0	Verified that all available reference and award amounts were in U.S. dollars; no currency conversion was required.
Zero or negative amounts	0	Verified absence before logarithmic transformations; no monetary correction was required.
Missing CPC in awards	0	Retained the original CPC and created cpc_5 from its first five digits solely for recurrence analysis.
Extreme values under the 1.5 x IQR rule	1,831 awards (11.65%)	Amounts were neither removed nor replaced. Original values were retained; log10 and 99th-percentile winsorized variables were created only for visualization.
Duplicate bidder participation	0	Verified the OCID + tenderer_id combination; no records were removed.
Duplicate awards	0	Verified the OCID + award_id + supplier_id combination; no records were removed.

Note. Own preparation based on SERCOP open data for 2025.

Annex 2 Dictionary of Variables Used in the Analysis

Table A2 Identification and description of model variables

Variable	Source or OCDS field	Type	Description	Role in analysis
ocid	release.ocid	String - nominal categorical	Unique identifier for each procurement procedure.	Identification and integration key.
buyer_id	buyer.id	String - nominal categorical	Identifier of the contracting entity used to recognize the buyer.	Identification - buyer node.

Variable	Source or OCDS field	Type	Description	Role in analysis
tenderer_id	tender.tenderers[].id	String - nominal categorical	Identifier of the bidder participating in the procedure.	Bidder identification and participation analysis.
supplier_id	awards[].suppliers[].id	String - nominal categorical	Identifier of the supplier receiving the award.	Identification - supplier node.
number_of_tenderers	tender.numberOfTenderers	Discrete quantitative	Number of bidders registered in each procedure.	Independent - participation or competition level.
tender_value	tender.lots[0].value.amount	Continuous quantitative	Procedure reference value in U.S. dollars.	Independent - economic reference value.
award_amount	awards[].value.amount; correctedValue.amount when required	Continuous quantitative	Final amount awarded to the supplier in U.S. dollars.	Independent - monetary basis for concentration and network analyses.
tender_start_date	tender.tenderPeriod.startDate	Temporal	Local procedure start date.	Temporal delimitation of procedures initiated in 2025.
award_date_local	awards[].date	Temporal	Local date on which the award was recorded.	Temporal and recurrence analysis.
cpc_5	Derived from awards[].items[].classification.id	String - nominal categorical	Five-digit CPC code used to compare similar procurement objects.	Classification for recurrence analysis.
numero_adjudicaciones	Derived from awards	Discrete quantitative	Total number of awards recorded per contracting entity.	Derived - entity activity level.
numero_proveedores	Derived from supplier_id	Discrete quantitative	Number of distinct suppliers awarded by each contracting entity.	Derived - supplier diversity.
hhi_monetario	Derived from award_amount	Continuous quantitative	HHI calculated from each supplier's share of total awarded amount.	Dependent - monetary concentration indicator.
hhi_frecuencia	Derived from award frequency	Continuous quantitative	HHI calculated from each supplier's share of award count.	Dependent - frequency concentration indicator.
hhi_monetario_normalizado	Derived from hhi_monetario and supplier count	Continuous quantitative	Normalized monetary HHI for comparing entities with different supplier structures; reported for entities with at least five awards of USD 5,000 or more and undefined when n = 1.	Dependent - normalized monetary concentration result.
hhi_frecuencia_normalizado	Derived from hhi_frecuencia and supplier count	Continuous quantitative	Normalized frequency HHI under the same eligibility conditions and undefined when n = 1.	Dependent - normalized frequency concentration result.
cr4_monetario	Derived from award_amount	Continuous quantitative	Share of total awarded amount concentrated among the four leading suppliers of each entity.	Dependent - complementary concentration indicator.
entropia_monetaria	Derived from award_amount	Continuous quantitative	Complementary diversity measure for the distribution of awarded amount among suppliers.	Dependent - diversity or concentration indicator.
max_procesos_90d	Derived from buyer_id, supplier_id, cpc_5, and award_date_local	Discrete quantitative	Maximum number of procedures for the same buyer, supplier, and CPC-5 within 90 days.	Dependent - recurrence intensity.
recurrencia_90d	Derived	Binary	Identifies buyer-supplier-CPC-5 combinations with more than three	Dependent - recurrence indicator.

Variable	Source or OCDS field	Type	Description	Role in analysis
			procedures in a 90-day window.	
frecuencia_relacion	Derived from buyer_id and supplier_id	Discrete quantitative	Number of awards accumulated between one contracting entity and one supplier.	Buyer-supplier relationship property.
monto_total_relacion	Derived from award_amount	Continuous quantitative	Total awarded amount accumulated between one contracting entity and one supplier.	Monetary weight of the relationship.
log10_monto_total_relacion	Derived from monto_total_relacion	Continuous quantitative	Base-10 logarithm of accumulated amount, used to reduce the visual effect of extreme values.	Transformed relationship weight in the network.
degree	Derived through Neo4j network analysis	Discrete quantitative	Number of direct connections of a contracting entity or supplier in the bipartite network.	Dependent - actor connectivity.
betweenness	Derived through Neo4j network analysis	Continuous quantitative	Betweenness measure identifying actors located on relevant network paths.	Dependent - actor betweenness position.
cr4_frecuencia	Derived from award frequency	Continuous quantitative	Share of awards concentrated among the four suppliers with the greatest frequency for each entity.	Dependent - complementary frequency concentration indicator.
entropia_frecuencia	Derived from award frequency	Continuous quantitative	Complementary diversity measure for the distribution of award frequency among suppliers.	Dependent - frequency diversity or concentration indicator.

Note. Own preparation based on SERCOP open data for 2025.

Annex 3 Complementary Visualizations Of The Analysis

Figure A1 Suppliers with the largest amount of accumulated awarded

Note. Own-production based on the open SERCOP data for 2025.

Figure A1 shows the ranking of suppliers with the highest accumulated amount awarded. To control the influence of the extreme values, winsorization was applied to the 99 percentile (P99 = USD 1,006.458.20) on the individual amounts before accumulating them per supplier, while the original values were retained for the other analyses. Only six of the ten suppliers coincided with the original ranking, confirming that the extreme values influenced its composition.

Figure A2 Suppliers with the highest number of awards

Note. Own-production based on the open SERCOP data for 2025.

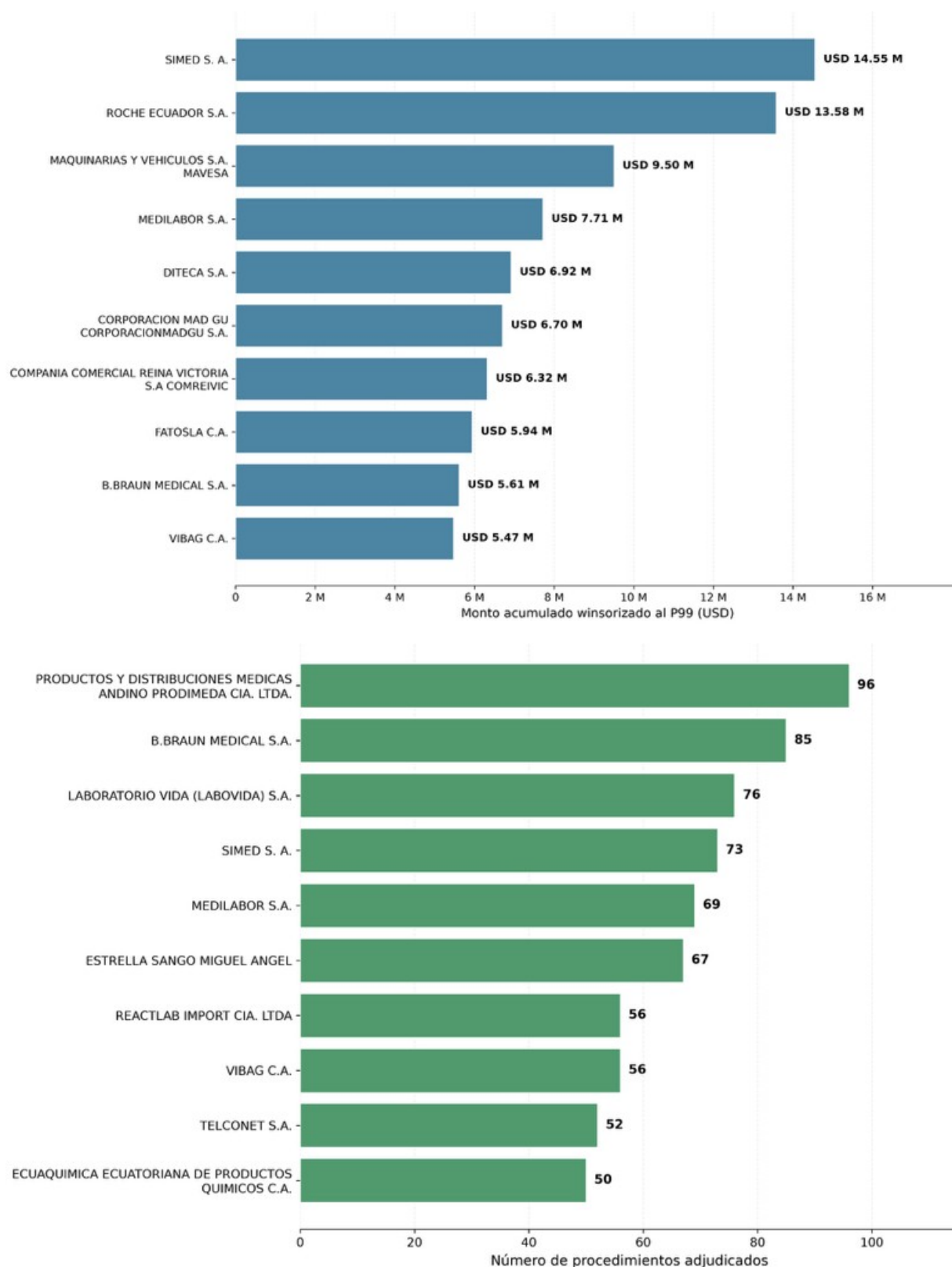


Figure A2 shows that Prodimedia Cia. Ltd. recorded the highest frequency, with 96 awards, followed by B. Braun Medical S.A., with 85, and Labovida S.A., with 76. Burgos Carlosama (2025) explains that a higher frequency does not necessarily imply a higher economic weight; therefore, this ranking identifies participation, but not concentration.

Figure A3 Entities with higher monetary concentration of awards

Note. Own-production based on the open SERCOP data for 2025.

Figure A3 shows that the Ministry of Production presented the highest standard monetary HHI (0.895), followed by District Directorate 08D04 (0.885) and Municipal GAD of Esmeraldas (0.845). According to Pliatsidis (2024), a high HHI must be interpreted as considering the market and is a sign to guide reviews, not direct evidence of collusion or irregularity.

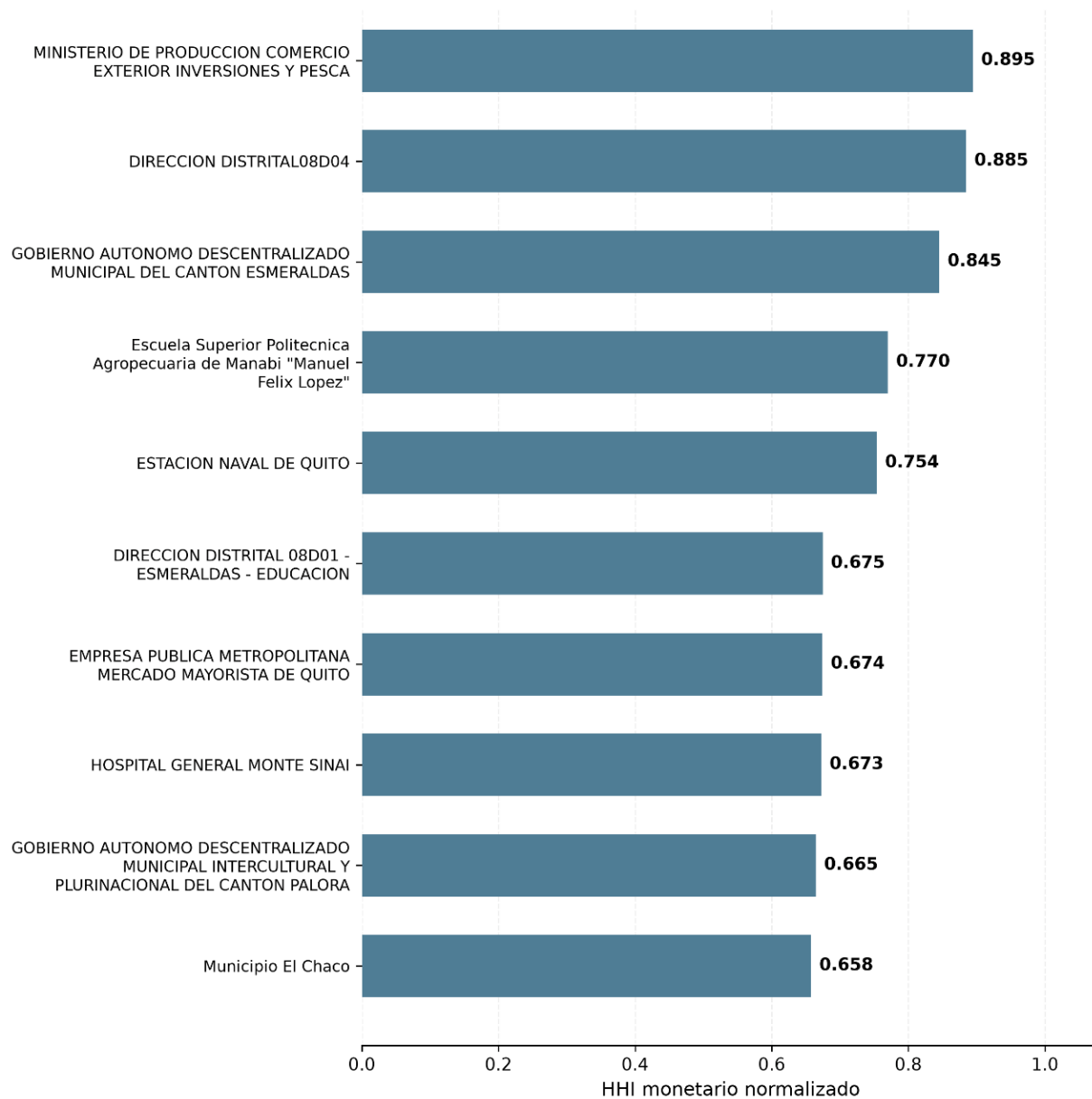
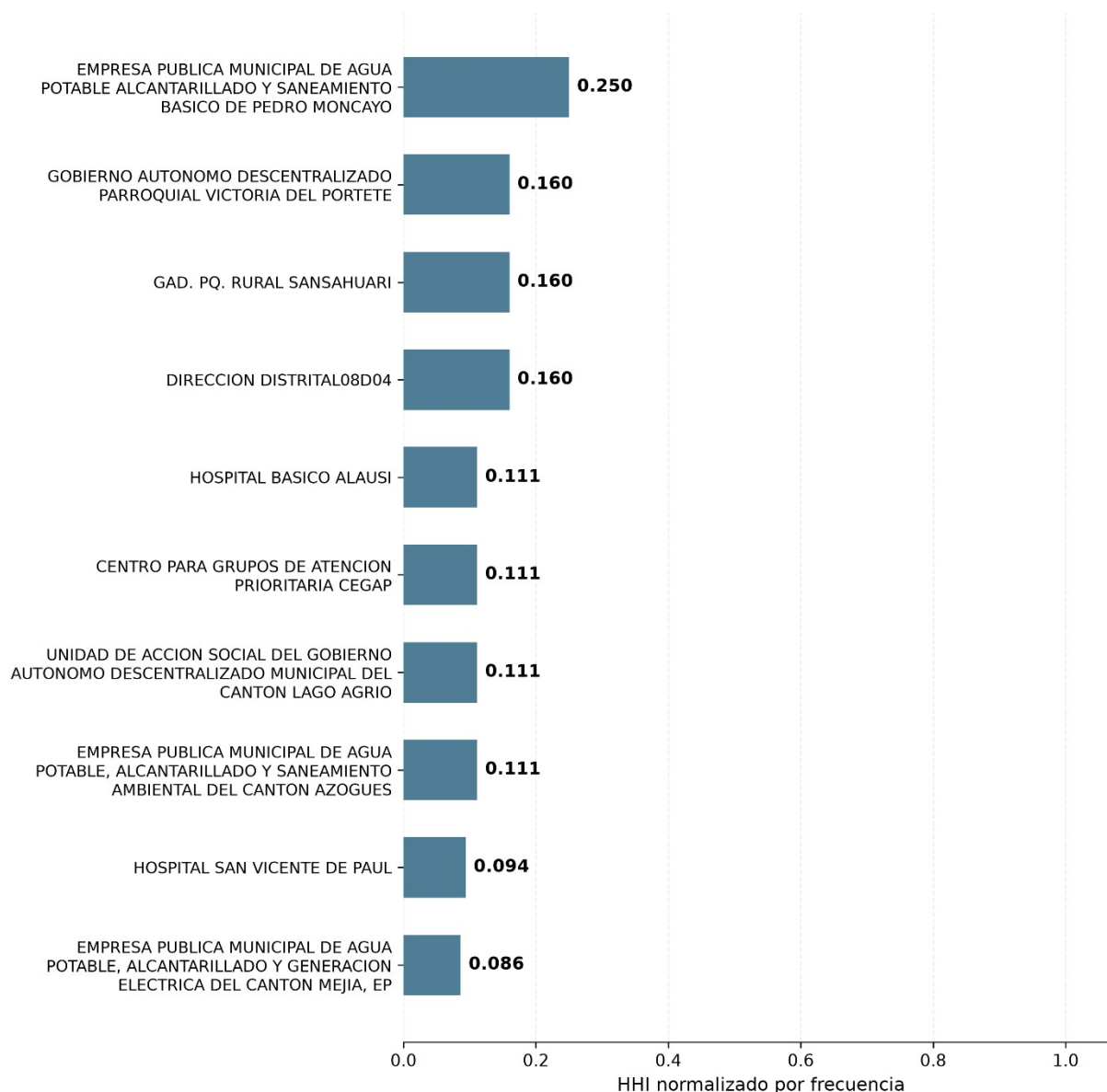


Figure A4 Entities with higher concentration by frequency of award

Note. Own-production based on the open SERCOP data for 2025.

Figure A4 shows that the Municipal Public Company of Drinking Water, Sewage and Basic Sanitation of Pedro Moncayo presented the highest HHI standard by frequency (0.250), with eight awards distributed among four suppliers. As Burgos Carlosama (2025) points out, the frequency and amount may present different concentration patterns and should be analysed separately.



Annex 4 Project Repository In Github

The complete technical development of the project is available in the following repository of GitHub:

<https://github.com/Ilchampo/ecuador-procurement-network-analytics>

The repository collects the data, codes and documents used during the different stages of the study. Its organization allows to maintain traceability from the original files published by the National Public Procurement Service to the processed bases and indicators obtained through Python and Neo4j.

The main content of the repository is organized as follows:

- Original data: the data/raw/2025 folder contains the twelve monthly JSON files downloaded from the SERCOP open data portal. These files were kept unchanged to preserve the integrity of the source.
- Processed bases: the data / processed folder contains the bases resulting from the integration, cleaning, standardization and validation processes. These relate to procedures, participation of bidders and awards, and include the derived variables and control flags used in the analysis.
- Notebooks of the project: the notebooks folder gathers the notebooks developed for the integration of the monthly files, the cleaning and validation of the data, the exploratory analysis, the construction of indicators, the generation of visualizations and the other analytical stages of the project.
- Analysis in Neo4j: the notebooks / neo4j folder contains the notebook 01 _ analysis _ red _ buyer _ supplier _ gds.ipb, related to the methodology described in point 4.5 of the report. This file documents the undirected construction and projection of the Buyer network

Supplier, as well as the application of the Weakly Connected Components (WCC), Degree, Betweenness, Louvain and Leiden algorithms.

The Neo4j notebook also incorporates Cypher consultations, execution parameters, validations and outputs used to analyze network components, actors' position, community detection, concentration, recurrence and signal combination. In this way, the repository complements the information presented in the report, avoids repeating extensive blocks of code and allows for the verification of the correspondence between data, methodology and results.

The availability of original files, processed bases and notebooks promotes the transparency and reproducibility of the project. The indicators obtained represent descriptive signals to guide review and monitoring processes, and do not in themselves constitute evidence of irregularity.

Project repository: <https://github.com/Ilchampo/ecuador-procurement-network-analytics>